

60016 OPERATIONS RESEARCH

Dual Simplex Algorithm

Program

Main threads seen in the course

- ▶ Simplex
- ▶ Duality

This lecture will join them

- ▶ Dual simplex

**In a nutshell: as applying simplex to dual program ...
... but directly operating on primal tableau**

Dual of the standard form primal

$$\begin{array}{ll} \min & c^T x \\ \text{s.t.} & Ax = b \\ & x \geq 0 \end{array} \quad \Longleftrightarrow \quad \begin{array}{ll} -\max & -c^T x \\ \text{s.t.} & Ax \leq b \\ & -Ax \leq -b \\ & x \geq 0 \end{array}$$

Dual program is

$$\begin{array}{ll} -\min & b^T z_1 - b^T z_2 \\ \text{s.t.} & A^T(z_1 - z_2) \geq -c \\ & z \geq 0 \end{array} \quad \Longleftrightarrow \quad \begin{array}{ll} \max & b^T y \\ \text{s.t.} & A^T y \leq c \end{array}$$

Dual simplex

The dual simplex is the simplex algorithm applied to the dual

$$\begin{array}{ll}\max & b^T y \\ \text{s.t.} & A^T y + w = c \\ & w \geq 0\end{array}$$

Simplex: enforce **primal feasibility** and step to primal optimality.

Dual simplex: enforce **dual feasibility** and step to dual optimality.

Q: what does the dual simplex correspond to on primals?

From primal BS to corresponding dual BS

Fix basis B for (P). We construct a corresponding basis $\hat{B}$ for (D).

$$A = \begin{bmatrix} B & N \end{bmatrix} \quad w = \begin{bmatrix} w_B \\ w_N \end{bmatrix}$$

Dual constraint $A^T y + w = c$ reads as $\begin{bmatrix} B^T \\ N^T \end{bmatrix} y + \begin{bmatrix} w_B \\ w_N \end{bmatrix} = c$, so

$$\begin{bmatrix} B^T & I & 0 \\ N^T & 0 & I \end{bmatrix} \begin{bmatrix} y \\ w_B \\ w_N \end{bmatrix} = c$$

Observation: the blue submatrix is squared. We have a dual basis

$$\begin{bmatrix} B^T & 0 \\ N^T & I \end{bmatrix} \begin{bmatrix} y \\ w_N \end{bmatrix} = c \quad w_B = 0$$

$$\hat{B} = \begin{bmatrix} B^T & 0 \\ N^T & I \end{bmatrix}, \quad \hat{B}^{-1} = \begin{bmatrix} B^{-T} & 0 \\ -N^T B^{-T} & I \end{bmatrix}$$

Primal dual basis properties

Obs 1: if $\hat{B}$ feasible for (D), then B optimal for (P).

Proof: $\hat{B}$ feasible for (D) implies $w_N \geq 0$, i.e.,

$$w_N = -N^T B^{-T} c_B + c_N \geq 0 \quad \Longleftrightarrow \quad r \geq 0$$

Obs 2: if $\hat{B}$ optimal for (D), then B feasible for (P).

Proof: $\hat{B}$ optimal for (D) implies dual red cost ≤ 0 (max).

$$0 - [I \quad 0] \begin{bmatrix} B^{-1} & -B^{-1}N \\ 0 & I \end{bmatrix} \begin{bmatrix} b \\ 0 \end{bmatrix} = -B^{-1}b \leq 0 \quad \Longleftrightarrow \quad \text{(P) feasible}$$

“dual simplex maintains dual feasibility, step to dual optimality”

=

“dual simplex maintains primal optim, step to primal feasibility”

Why is it useful?

- ▶ if already solved problem, and need to resolve for different constraints, reuse initial BFS for dual (i.e., no need to run phase I again!)
- ▶ on average faster than primal LP for typical problems

How to run dual simplex on primal tableau

All steps are identical with the exception of

- ▶ **initial point:** start from point ensuring *optimality*
- ▶ **variable out:** row corresponding most negative *RHS*
- ▶ **variable in:** ratio test over *columns*
- ▶ **halting test:** we now check for *feasibility*

Example

$$\begin{array}{ll}\max & -4x_1 - 2x_2 - x_3 \\ \text{s.t.} & -x_1 - x_2 + 2x_3 \leq -3 \\ & -4x_1 - 2x_2 + x_3 \leq -4 \\ & x_1 + x_2 - 4x_3 \leq 2 \\ & x \geq 0\end{array}$$

$$\begin{array}{ll}\min & -3y_1 - 4y_2 + 2y_3 \\ \text{s.t.} & -y_1 - 4y_2 + y_3 \geq -4 \\ & -y_1 - 2y_2 + y_3 \geq -2 \\ & 2y_1 + y_2 - 4y_3 \geq 1 \\ & y \geq 0\end{array}$$

BV	z	x ₁	x ₂	x ₃	x ₄	x ₅	x ₆	RHS	
z	1	-4	-2	-1	0	0	0	0	
x ₄	0	-1	-1	2	1	0	0	-3	dual obj coeff
x ₅	0	-4	-2	1	0	1	0	-4	
x ₆	0	1	1	-4	0	0	1	2	

We have basic solution (infeasible). But it is feasible for dual (see row of reduced cost ≤ 0), and corresponds to $y = 0$

Example cont.

Move towards primal feasibility

BV	z	x_1	x_2	x_3	x_4	x_5	x_6	RHS
z	1	-4	-2	-1	0	0	0	0
x_4	0	-1	-1	2	1	0	0	-3
x_5	0	-4	-4	-2	1	0	1	-4
x_6	0	1	1	-4	0	0	1	2

← pivot row

Increasing y_2 decreases dual objective $\implies$ pivot on row 3

By how much can we increase y_2 before hitting a constraint?

$$-y_1 - 4y_2 + y_3 \geq -4 \implies y_2 \leq 4/4$$

$$-y_1 - 2y_2 + y_3 \geq -2 \implies y_2 \leq 2/2$$

$$2y_1 + y_2 - 4y_3 \geq -1 \implies \text{no constraint}$$

Here equivalent pivot on column for x_1 or x_2 , pick x_1

Concluding this part

Recap

- ▶ Simplex, stopping and degeneracies
- ▶ Duality
- ▶ Sensitivity and shadow prices